

Sadeem Mehkery

UAE resident | sadeem.meh10@gmail.com | 050 854 2136 | LinkedIn: Sadeem Mehkery | GitHub: sadreammm

EDUCATION

University of Wollongong in Dubai

Dubai, UAE

Bachelor of Computer Science (Artificial Intelligence & Big Data), GPA: 4.0

Expected Graduation - 2026

- **Relevant Coursework:** Object Oriented Programming, Database Systems, Data Structures & Algorithms, Computer Vision, Machine Learning, Big Data Management, C++ Programming

EXPERIENCE

AI Intern

Dubai, UAE

KSB

August 2025 - September 2025

- Built a scalable FastAPI server with REST APIs, JWT authentication, ORM storage, and OCR/PDF parsing, automating specification intake and boosting throughput 5x.
- Developed a semantic retrieval and pricing matcher by fine-tuning MPNet embeddings on specification data; deployed API endpoints with human-in-the-loop verification for accurate extraction
- Automated the RFQ pipeline (JWT-protected intake, duplicate detection via hashing, structured parsing, JSON normalization), reducing manual entry by 50% and generating cleaner training data for AI models.
- Deployed the backend with CI/CD and delivered a React frontend for quoting workflows, cutting offer preparation time by 60% and accelerating iteration on ML-driven recommendations.

PROJECTS

OptiRoute – AI-Powered Taxi Allocation | Scikit-learn, Statsmodels, Prophet, Seaborn, Matplotlib, Plotly

- Optimized fleet allocation for Dubai ride-hailing services using AI-driven demand forecasting and allocation models.
- Contributed to building zone-level forecasting models (Prophet, ARIMA, Decision Trees) and designed uncertainty-aware allocation heuristics, reducing idle variance by 25–40%.
- Implemented data preprocessing, feature engineering (temporal & spatial via K-Means), and custom KPIs (Match Rate, Fuel Waste, Unmet Demand).
- Developed visual dashboards (Seaborn, Matplotlib, Plotly) to benchmark models and simulate short-term (12h) and long-term (5d) fleet planning.

OnStack - AI-Powered Intelligent Enterprise System | Fastapi, React.js, RL, RAG, NLP

- Reduced employee onboarding time by 30% by building an AI-driven guidance system with real-time contextual overlays that walks new hires through platform workflows with interactive tutorials.
- Achieved 87% guidance accuracy through reinforcement learning that optimizes instruction delivery based on user feedback signals, automatically adapting guidance complexity to individual learning patterns.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with a YAML-based knowledge base to ground Gemini responses in internal documentation, enabling 90%+ confidence natural-language task parsing and workflow generation.
- Built a knowledge preservation system that captures institutional knowledge during employee exits and generates handover playbooks, reducing knowledge loss risk by 40% through federated learning.

Enterprise-Scale Analytics (EY Case Study) | Hadoop, Spark, Hive, Kafka, Power BI

- Built a scalable Hadoop & Spark data platform processing multi-format enterprise data using ETL pipelines with HDFS, Hive, Kafka, and MapReduce for enterprise analytics.
- Developed batch and real-time ETL workflows, including Spark-based ML risk scoring applied to streaming CRM data via Kafka.
- Enabled secure, analytics-ready datasets through data cleansing, schema normalization, and role-based access control (RBAC).
- Delivered SQL-accessible analytical views and Power BI dashboards to support finance, HR, audit, and operational reporting.

SKILLS

Programming Languages: Python, SQL, HiveQL, JavaScript, C/C++

Big Data & Data Engineering: Hadoop (HDFS, MapReduce), Spark (SQL, Streaming, ML), Hive, Kafka, HBase

Frameworks: React, Node.js, Express, Flask, FastAPI

Libraries: TensorFlow, Scikit-learn, OpenCV, Langchain, PySpark

Databases: MySQL, PostgreSQL, MongoDB, Pinecone, Neo4j

Tools: GitHub, AWS, Postman, HuggingFace, Docker, Power BI, Linux